(a) $(Ax+B)e^{2x}$

(b) $(Ax+B)e^{-2x}$

(c) $A\cos 2x + B\sin 2x$

(d) $Ae^{-2x} + Be^{2x}$

9. The differential equation of $y = mx + c$ is (m and c are arbitrary constants)

(a) $\frac{d^2y}{dx^2} = o$ (b) $y = x\frac{dy}{dx} + c$

(c) $xdy + ydx = 0$ (d) $ydx - xdy = 0$

10. The particular integral of the differential equation is $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 2e^{4x}$

(a) $\frac{x^2e^{4x}}{2!}$ (b) $\frac{e^{4x}}{2!}$

(c) x^2e^{4x} (d) xe^{4x}

11. Solution of $\frac{dx}{dy} + Px = 0$

(a) $x = ce^{py}$ (b) $x = ce^{-py}$

(c) $x = py + c$ (d) $x = cy$

12. If $\sec^2 x$ is an integrating factor of the differential equation $\frac{dy}{dx} + Py = Q$ then $P =$

(a) $2\tan x$ (b) $\sec x$

(c) $\cos^2 x$ (d) $\tan^2 x$

13. The integrating factor of $x\frac{dy}{dx} - y = x^2$ is

(a) $\frac{-1}{x}$ (b) $\frac{1}{x}$

(c) $\log x$ (d) x

14. The solution of the differential equation $\frac{dy}{dx} + Py = Q$ where P and Q are the function of x is

(a) $y = \int Qe^{\int Pdx}dx + c$

(b) $y = \int Qe^{-\int Pdx}dx + c$

(c) $ye^{\int Pdx} = \int Qe^{\int Pdx}dx + c$

(d) $ye^{\int Pdx} = \int Qe^{-\int Pdx}dx + C$

15. The differential equation formed by eliminating A and B from $y = e^{-2x}(A\cos x + B\sin x)$ is

(a) $y_2 - 4y_1 + 5 = 0$

(b) $y_2 + 4y - 5 = 0$

(c) $y_2 - 4y_1 - 5 = 0$

(d) $y_2 + 4y_1 + 5 = 0$

16. The particular integral of the differential equation $f(D)y = e^{ax}$ where $f(D) = (D-a)^2$

(a) $\frac{x^2}{2}e^{ax}$ (b) xe^{ax}

(c) $\frac{x}{2}e^{ax}$ (d) x^2e^{ax}

17. The differential equation of $x^2 + y^2 = a^2$

(a) $xdy+ydx=0$ (b) $ydx-xdy=0$

(c) $xdx-ydx=0$ (d) $xdx+ydy=0$

18. The complementary function of $\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$ is

(a) $A + Be^x$ (b) $(A+B)e^x$

(c) $(Ax+B)e^x$ (d) $Ae^x + B$

19. The P.I of $(3D^2 + D - 14)y = 13e^{2x}$ is

(a) $\frac{x}{2}e^{2x}$ (b) xe^{2x}

(c) $\frac{x^2}{2}e^{2x}$ (d) $13xe^{2x}$

20. The general solution of the differential equation $\frac{dy}{dx} = \cos x$ is

(a) $y = \sin x + 1$ (b) $y = \sin x - 2$

(c) $y = \cos x + c$, c is an arbitrary constant

(d) $y = \sin x + c$, c is an arbitrary constant

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